

Reasoning At Play, Simplify Anyway

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GAP 12

Düsseldorf, 10 September, 2025

IN SHORT: This presentation concerns the question of whether some doxastic states, in virtue of their alleged involuntary nature, are unsuitable to play simplification roles in our reasoning. I argue that simplification roles need to be assessed in terms causal efficacy of attitudes, not in terms of control profiles.

1. How to Make it Simple?

An Illustration:

Case 1 (Planning). In planning my day—a June day in Palo Alto—I simply take it for granted that it will not rain even though I am not certain about this. If I were instead figuring out at what odds I would accept a monetary bet from you on the weather I would not simply take it for granted that it will not rain. But in my present circumstances taking this for granted simplifies my planning in a way that is useful, given my limited resources for reasoning. (Bratman 1992, p. 5)

To make our reasoning simple enough is crucial for coping with a world with lots of information. If we don't do it, our reasoning threaten to become intractably complex.

We simplify reasoning when we reason using a subset of the information available. Sometimes this is done by default, sometimes as a result of environmental pressures, and sometimes we do it with the explicit intention to simplify.

How do we make our reasoning simple enough?

CONSIDER an approach that uses the idea of **reliant doxastic states**:

An agent A reliantly is in a doxastic state $\mathcal{D}$ towards a proposition p just in case A is disposed to reason by assuming p .

The right kind of reliant doxastic state, in the sense above, helps to make our reasoning simpler than it would otherwise be.

In the literature on simplified reasoning,² the following necessary condition has been proposed on a doxastic state playing the simplification role:

Simplifying Role $\approx$ A doxastic state $\mathcal{D}p$ plays the simplification role only if $\mathcal{D}p$ disposes A to assume p in reasoning.³

Belief, arguably, plays such a role. When an agent A believes that p , A is thereby disposed to assume p in reasoning, a disposition that has a simplifying effect in the agent's reasoning.⁴

Notice that $\mathcal{D}$, itself, can be closely linked to syntactically richer states. A credence $c(p) = x$ can help to simplify in virtue of a heuristic bridge between credence and assumption.⁵

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Thanks for your feedback!

Plan:

1. How to make it simple?
2. The Discussion
3. A Controlled Automaticity View

Let's stipulate here that the agent does *not believe* that it won't rain (*pace* Bratman).

Bratman-Acceptance $\approx$ an inner state on the basis of which we form and maintain plans and intentions and that is sensitive to different practical pressures (e.g., the necessity to simplify, high stakes, cooperation, special obligations, among other things.) See Soter 2023 for a recent treatment.

Belief $\approx$ an inner state that is context-independent, responsive to evidence and truth-directed, involuntary (i.e. not formed by choice), and under rational requirements of coherence (see Williams 1970). In the probabilistic case, credences are accuracy-directed.

² Ross and Schroeder 2014.

³ The background **intuition**: unless A assumes p in reasoning, A must represent possibilities without ruling out $\neg p$. If so, more complex reasoning rules need to be involved.

⁴ **But**: see for discussion Wimmer (2021).

⁵ See Tang 2015; Dinges 2022. For discussion, see Palmira 2023.

TWO MAIN questions in the literature:

A. Rationality Question:⁶ if more information leads to more knowledge and better decisions, how and why is simplification rational? Is simplification epistemically or practically rational?

B. State-Functional Question: Which doxastic state(s) $\mathcal{D}$ is(are) able to play the simplification role?⁷

THINK about the latter: maybe there are certain $\mathcal{D}$'s which, by their very nature, are unable to play a simplification role. Those $\mathcal{D}$'s, perhaps, play causal roles in reasoning in such a way that is implausible to say they do interesting simplification jobs. Others, perhaps, are better suited for the job.

AS PROMISED, I will propose an answer to the question: do control features of doxastic states matter for the purposes of answering the **State-Functional Question**?

2. The Discussion

Some motivation for why this question is interesting: consider part of the discussion in Dinges (2021).

In the context of the question of whether simplified reasoning helps to justify the inclusion of belief in our ontology of mental states, he defends that **no**, it doesn't.

The issue is to explain what psychological states help us to assume things in reasoning.

Dinges argues we can explain that with credences. When we are in states of high enough $c(p) = x$, we can be disposed assume p in reasoning.

If that's right, we don't need to identify assumption and belief in a sense relevant for simplification.

In fact, according to Dinges, it's implausible to hold that. He says:

what we treat as true is under our voluntary control in a way that belief is not. I can decide to consider the possibility of rain in planning my day in Palo Alto and thus stop treating it as true that it will not rain. Staffel (2019, n.5) states, contrarily, that 'we usually can't employ deliberative control over which claims we take for granted in framing a reasoning problem, this is done automatically and without our conscious awareness'. I agree that we often grant propositions 'automatically and without our conscious awareness' [...]. But it does not follow that we lack 'deliberative control'. Analogously, we often blink automatically without our conscious awareness. Nevertheless we can control our blinking if we want. (Dinges 2021, p. 3)

SUCCINCTLY PUT, for a doxastic state to play the simplification role, it crucially matters whether we can control it in the right way.

The following argument emerges:

P1. Simplification Flexibility: Reasoners can control directly and voluntarily when and how they simplify their reasoning.

P2. Simplification Voluntarism: If a doxastic state $\mathcal{D}$ plays a simplification role, then, necessarily, $\mathcal{D}$ is potentially under our direct voluntary control.

P3. Belief Involuntarism: Belief is an involuntary doxastic state.

$\therefore$ Belief can't play the simplification role.

⁶ See: Harsanyi 1985, Bratman 1992, Lance 1995, Holton 2008, Wedgwood 2012, Ross and Schroeder 2014, Staffel 2019, Dinges 2021, Palmira 2023.

⁷ Answers in the menu: $\mathcal{D}$ = Belief (Staffel 2019, Ross and Schroeder 2014), Credence (Dinges 2021), Acceptance (Dinges 2022), Imagining (Palmira 2023). Question B is mostly relevant in the context of "Bayesian Challenge" (see Jeffrey 1970 and Kaplan 1998, Ch. 4) is central in this discussion.

See Weisberg (2020).

High enough = $c(p) = x$, $x \geq t$, where t represents a contextually appropriate threshold for assumption.

3. A Controlled Automaticity View

Anticipating, I will present an argument to reject **P2**.

Let me begin by granting that cases such as **Planning** help to motivate **Simplification Flexibility**.

In fact, it is very plausible that, by its very nature, simplification necessitates flexibility.

IN ADDITION, although **P3** rings controversial to my ears, I will concede it for present purposes.

LET ME begin my argument for rejecting **P2** by working out a distinction between

Task Simplification: Consider again the **Planning** case. The agent makes reasoning simpler by assuming that it won't rain, where he could otherwise have not assumed that, and reasoned by, say, looking up rain probabilities. Arguably, to make reasoning tractable at all, we always engage in some form of task simplification, even when our reasoning is relatively complex.

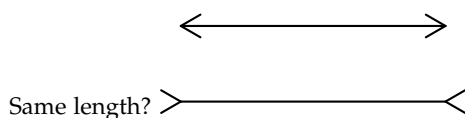
Simplified reasoning: This is just good-old reasoning facilitated by a suitable, reliant doxastic state that operates with a simplifying effect.

My contention will be that reliant doxastic states may **result** from task simplification.

But playing the simplification role is just about a reliant doxastic state's causal relations to other states in a temporally extended reasoning sequence.

When we arrive at a solution to a reasoning task by simplified reasoning, it won't matter how **task simplification** came about.

Consider Dan, a character who first encounters the Müller-Lyer illusion:



Dan's impression is that the lower line is longer. Dan doesn't measure the lines, and believes as a result that the lower line is longer.

Dan's cognitive mechanisms involved in size perception make the illusion persist, a persistence suggestive of lack of control of those mechanisms, which result in a correspondingly persistent belief.

What the mechanisms involved in size perception do, in the context of size judgements, is to make the processing of size information simpler. This is task simplification.

Notice: we lack control of the mechanisms involved in size perception that result in the Müller-Lyer illusion, if we lack control of anything in our cognitive apparatus.

But we wouldn't say that Dan can't decide to stop assuming his wrong conclusions on the basis of that lack of control. In fact, we can help him change his wrong conclusions:

Consider: cases of high environmental pressure, like little time to answer a question or a bear chasing you.

We approach these cases by presupposing that agents can exert direct voluntary control over the way they reason.

Little time: take your best guess (Dorst and Mandelkern 2021).

Bear: climb the nearest tree that seems high enough! (Lin 2014)

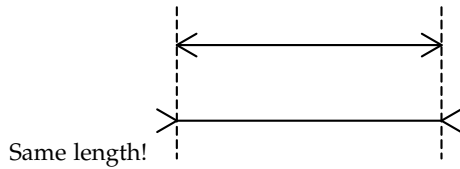
Two different focal points to assess the possibility of control.

MY SUGGESTION:

We exert control over task simplification, but not over simplified reasoning (more below).

Hence my slogan:

When reasoning is at play, we simplify anyway!



SOMETHING SIMILAR HAPPENS in the [Planning](#) case.

When our agents switches from reasoning for planning to reasoning for betting, we reveal a kind of control that is typically present in *tokens* of reasoning.

In reasoning processes where task simplification happens, it is not necessarily the case that we arrive to conclusions using reliant doxastic states which—like belief and perceptual seemings—are beyond our direct and voluntary control.

If reliant doxastic states are causally effective in reasoning, the fact that we can't control them doesn't speak against their suitability.

In summary, the following is the way I suggest we reject **P2**:

We don't need to deny the possibility of controlling what we assume in reasoning.

Rather, when we are operating under a particular assumption, reasoning is already at play. Here, it is not up to us what conclusions we will reach.

Whether or not one defends **P3**, simplification flexibility is compatible with belief.

Wrap Up

First, I motivated my question by showing how simplified reasoning, its flexibility, and direct voluntary control might be connected.

Afterwards, I revealed problematic to propose some necessary link between simplified reasoning and voluntary control.

Lastly, I developed a view of controlled automaticity. We can control how we reason and how to characterize our reasoning tasks, but when reasoning is at play, it will take its automatic course.

Belief involuntarism, the thesis that it is impossible to directly and voluntarily control what we believe, may find some support in how I picture simplified reasoning.

Maybe the idea that there are stages where we must give up control in order to put reasoning at play can support the intuitive pull of involuntarism.

Maybe the compatibility of simplification flexibility and belief involuntarism also shows that we don't need to solve problems of doxastic voluntarism in order to provide a working theory of simplified reasoning.

BUT, does involuntarism remain an interesting thesis? If we can exert control with doxastic effects, maybe involuntarism loses some of its dialectical grip in larger debates.

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